

Table 1 Microeconomic Model Calibration

Calibrated Parameters			
Description	Parameter	Value	Source
Permanent Income Growth Factor	$\mathcal{G}$	1.03	PSID: Carroll (1992)
Interest Factor	R	1.04	Conventional
Time Preference Factor	β	0.96	Conventional
Coefficient of Relative Risk Aversion	γ	2	Conventional
Probability of Zero Income	$\wp$	0.005	PSID: Carroll (1992)
Std Dev of Log Permanent Shock	σ_ψ	0.1	PSID: Carroll (1992)
Std Dev of Log Transitory Shock	σ_θ	0.1	PSID: Carroll (1992)

{table:Parameters}

Table 2 Model Characteristics Calculated from Parameters

Description	Symbol and Formula		Approximate Calculated Value
Finite Human Wealth Factor	$\tilde{\mathcal{R}}^{-1}$	$\equiv \mathcal{G}/R$	0.990
PF Value of Autarky Factor	$\sqsupset$	$\equiv \beta \mathcal{G}^{1-\gamma}$	0.932
Growth Compensated Permanent Shock	$\underline{\psi}$	$\equiv (\mathbb{E}[\psi^{-1}])^{-1}$	0.990
Uncertainty-Adjusted Growth	$\underline{\mathcal{G}}$	$\equiv \mathcal{G} \underline{\psi}$	1.020
Utility Compensated Permanent Shock	$\underline{\underline{\psi}}$	$\equiv (\mathbb{E}[\psi^{1-\gamma}])^{1/(1-\gamma)}$	0.990
Utility Compensated Growth	$\underline{\underline{\mathcal{G}}}_u$	$\equiv \mathcal{G} \underline{\underline{\psi}}$	1.020
Absolute Patience Factor	$\mathfrak{P}$	$\equiv (R\beta)^{1/\gamma}$	0.999
Return Patience Factor	$\mathfrak{P}/R$	$\equiv \mathfrak{P}/R$	0.961
Growth Patience Factor	$\mathfrak{P}/\mathcal{G}$	$\equiv \mathfrak{P}/\mathcal{G}$	0.970
Modified Growth Patience Factor	$\mathfrak{P}/\mathcal{G}\mathbb{E}[\psi^{-1}]$	$\equiv \mathfrak{P}/\underline{\mathcal{G}}$	0.980
Value of Autarky Factor	$\sqsupseteq$	$\equiv \beta \mathcal{G}^{1-\gamma} \underline{\underline{\psi}}^{1-\gamma}$	0.941
Weak Return Impatience Factor	$\wp^{1/\gamma} \mathfrak{P}$	$\equiv (\wp \beta R)^{1/\gamma}$	0.071

{table:Calibration}

Table 3 Definitions and Comparisons of Patience Conditions

Perfect Foresight Versions	Uncertainty Versions
Finite Limiting Human Wealth (FHWL)	
$\mathcal{G}/R < 1$ The growth factor for permanent income $\mathcal{G}$ must be smaller than the discounting factor R for human wealth to be finite.	$\mathcal{G}/R < 1$ The model's risks are mean-preserving spreads, so the PDV of future income is unchanged by their introduction.
Absolute Impatience Condition	
$\mathbf{P} < 1$ The unconstrained consumer is sufficiently impatient that the level of consumption will be declining over time: $\mathbf{c}_{t+1} < \mathbf{c}_t$	$\mathbf{P} < 1$ <i>If wealth is large enough, the expectation of consumption next period will be smaller than this period's consumption:</i> $\lim_{m_t \rightarrow \infty} \mathbb{E}_t[\mathbf{c}_{t+1}] < \mathbf{c}_t$
Return Impatience	
Return Impatience Condition (RIC)	Weak RIC (WRIC)
$\mathbf{P}/R < 1$ The growth factor for consumption $\mathbf{P}$ must be smaller than the discounting factor R , so that the PDV of current and future consumption will be finite: $c'(m) = 1 - \mathbf{P}/R < 1$	$\wp^{1/\gamma} \mathbf{P}/R < 1$ If the probability of the zero-income event is $\wp = 1$ then income is always zero and the condition becomes identical to the RIC. Otherwise, weaker. $c'(m) < 1 - \wp^{1/\gamma} \mathbf{P}/R < 1$
Growth Impatience	
Growth Impatience Condition (GIC)	Strong Growth Impatience (GIC-Mod)
$\mathbf{P}/\mathcal{G} < 1$ For an unconstrained PF consumer, the ratio of $\mathbf{c}$ to $\mathbf{p}$ will fall over time. For constrained, guarantees the constraint eventually binds. Guarantees $\lim_{m_t \uparrow \infty} \mathbb{E}_t[\psi_{t+1} m_{t+1}/m_t] = \mathbf{P}/\mathcal{G}$	$\mathbf{P}\mathbb{E}[\psi^{-1}]/\mathcal{G} < 1$ By Jensen's inequality stronger than GIC. Ensures consumers will not expect to accumulate m unboundedly. $\lim_{m_t \rightarrow \infty} \mathbb{E}_t[m_{t+1}/m_t] = \mathbf{P}/\mathcal{G}\mathbb{E}[\psi^{-1}]$
Finite Value of Autarky	
PFFVAC	FVAC
$\beta \mathcal{G}^{1-\gamma} < 1$ equivalently $\mathbf{P} < R^{1/\gamma} \mathcal{G}^{1-1/\gamma}$ The discounted utility of constrained consumers who spend their permanent income each period should be finite.	$\beta \mathcal{G}^{1-\gamma} \mathbb{E}[\psi^{1-\gamma}] < 1$ By Jensen's inequality, stronger than the PFFVAC because for $\gamma > 1$ and nondegenerate ψ , $\mathbb{E}[\psi^{1-\gamma}] > 1$.

{table:Comparison}

Table 4 Sufficient Conditions for Nondegenerate[‡] Solution

Consumption Model(s)	Conditions	Comments
$\bar{c}(m)$: PF Unconstrained $\underline{c}(m) = \underline{\kappa}m$ Section 2.3.1: Section 2.3.1: Eq (54) in Appendix A.2: Eq (53) in Appendix A.2:	RIC, FHWC [°]	RIC $\Rightarrow v(m) < \infty$; FHWC $\Rightarrow 0 < v(m) $ PF model with no human wealth ($h = 0$) RIC prevents $\bar{c}(m) = \underline{c}(m) = 0$ FHWC prevents $\bar{c}(m) = \infty$ PFFVAC+FHWC $\Rightarrow$ RIC GIC+FHWC $\Rightarrow$ PFFVAC
$\dot{c}(m)$: PF Constrained Section 5.1.1: Appendix D: Appendix D:	$\mathcal{GIC}$, RIC $\mathcal{GIC}$, RIC $\mathcal{GIC}$, $\mathcal{RIC}$	FHWC holds ($\mathcal{G} < \mathbf{D} < \mathbf{R} \Rightarrow \mathcal{G} < \mathbf{R}$) $\dot{c}(m) = \bar{c}(m)$ for $m > m_{\#} < 1$ ($\mathcal{RIC}$ would yield $m_{\#} = 0$ so $\dot{c}(m) = 0$) $\lim_{m \rightarrow \infty} \dot{c}(m) = \bar{c}(m)$, $\lim_{m \rightarrow \infty} \dot{\kappa}(m) = \underline{\kappa}$ kinks where horizon to $b = 0$ changes* $\lim_{m \rightarrow \infty} \dot{\kappa}(m) = 0$ kinks where horizon to $b = 0$ changes*
$c(m)$: Friedman/Muth Section 2.4.2: Section 5.2: Figure 8: Section 5.2.1: Case 3 Section 5.2.1: Case 1 Section 3.1: Theorem 3: Theorem 4:	Section 2.4.1 & 2.4.2 $\mathcal{FVAC}$, $\mathcal{WRIC}$	$\underline{c}(m) < c(m) < \bar{c}(m)$ $\underline{v}(m) < v(m) < \bar{v}(m)$ Sufficient for Contraction $\mathcal{WRIC}$ is weaker than RIC $\mathcal{FVAC}$ is stronger than PFFVAC $\mathcal{FHWC} + \mathcal{RIC} \Rightarrow \mathcal{GIC}$, $\lim_{m \rightarrow \infty} \kappa(m) = \underline{\kappa}$ $\mathcal{RIC} \Rightarrow \mathcal{FHWC}$, $\lim_{m \rightarrow \infty} \kappa(m) = 0$ “Buffer Stock Saving” Conditions $\mathcal{GIC} \Rightarrow \exists \tilde{m} \text{ s.t. } 0 < \tilde{m} < \infty$ $\mathcal{GIC}\text{-Mod} \Rightarrow \exists \hat{m} \text{ s.t. } 0 < \hat{m} < \infty$

[‡]For feasible m satisfying $0 < m < \infty$, a nondegenerate limiting consumption function defines a unique optimal value of c satisfying $0 < c(m) < \infty$; a nondegenerate limiting value function defines a corresponding unique value of $-\infty < v(m) < 0$.

[°]RIC, FHWC are necessary as well as sufficient for the perfect foresight case. *That is, the first kink point in $c(m)$ is $m_{\#}$ s.t. for $m < m_{\#}$ the constraint will bind now, while for $m > m_{\#}$ the constraint will bind one period in the future. The second kink point corresponds to the m where the constraint will bind two periods in the future, etc.

**In the Friedman/Muth model, the RIC+FHWC are sufficient, but *not* necessary for nondegeneracy

Table 5 Appendix: Perfect Foresight Liquidity Constrained Taxonomy

For constrained $\dot{c}$ and unconstrained $\bar{c}$ consumption functions

{table:LiqConstrSec

Main Condition Subcondition	Math	Outcome, Comments or Results
GIC and RIC	$1 < \mathbf{P}/\mathcal{G}$ $\mathbf{P}/R < 1$	Constraint never binds for $m \geq 1$ FWC holds ($R > \mathcal{G}$); $\dot{c}(m) = \bar{c}(m)$ for $m \geq 1$
and RIC	$1 < \mathbf{P}/R$	$\dot{c}(m)$ is degenerate: $\dot{c}(m) = 0$
GIC and RIC	$\mathbf{P}/\mathcal{G} < 1$ $\mathbf{P}/R < 1$	Constraint binds in finite time $\forall m$ FWC may or may not hold $\lim_{m \uparrow \infty} \bar{c}(m) - \dot{c}(m) = 0$ $\lim_{m \uparrow \infty} \dot{\kappa}(m) = \underline{\kappa}$
and RIC	$1 < \mathbf{P}/R$	FWC $\lim_{m \uparrow \infty} \dot{\kappa}(m) = 0$

Conditions are applied from left to right; for example, the second row indicates conclusions in the case where ~~GIC~~ and RIC both hold, while the third row indicates that when the GIC and the RIC both fail, the consumption function is degenerate; the next row indicates that whenever the GIC holds, the constraint will bind in finite time.